Lab instructions

Goal

In this Lab you'll explore the concept of keys in more depth, so you have a better understanding of how to choose a primary key from a list of "candidate keys", and how to connect tables together with foreign keys in SQL.

❖ Task 1

Scenario

Review the following vehicle table.

Vehicle table

Vehicle ID	Owner ID	Car plate number	Owner phone number
D01	Ow01	PL02NY	0738297294
D02	Ow02	SN02L2	0725021582
D03	Ow03	PK02L2	0765021583

1. Can you identify the candidate key attributes?

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2. You might notice that there are unique values in all rows that belong to the owner ID. So, is the owner ID a fourth candidate key?

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3. But which of the three attributes would be the best candidate to act as a primary key?

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4. Complete the following step-by-step instructions inside MySQL in your machine:
   a. Create the database **Automobile.**
   b. Create the **Vehicle** table inside the Automobile database.
   c. Show the structure of the **Vehicle** table.

5. Now any candidate key that has not been chosen to act as the primary key of the **Vehicle** table is called an alternate key. That's the plate number and phone number. Let's build another table to maintain data about the vehicles' owners. This table includes information about the owner's name and address as illustrated below.

Owner table

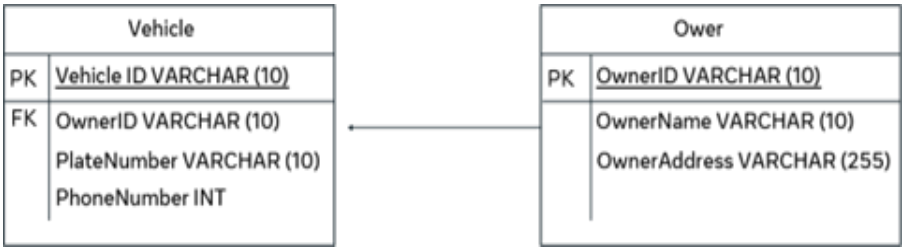
Owner ID	Owner name	Owner address
Ow01	Amjad Omer	110, Elephant Way
Ow02	Hans Henderson	120, Dragon Way
Ow03	Paulo Galdames	130, Giraffe Avenue

The owner ID is the primary key in this table, as it will always be unique. Whereas the owner's name could be the same for different owners, and the owner address could be the same for different owners living at the same address. This would violate the uniqueness requirement for the primary key.

a. Create the **Owner** table in the automobile database. Remember to declare the ownerID as the primary key of the table.
b. Examine the structure of the Owner table.

By now you have two tables created in the automobile database. The vehicle table contains information about the vehicle ID, owner ID, plate number and phone number. Whereas the owner table contains information about the owner ID, owner name and address.

6. You may have noticed that the owner ID is a common attribute, as it exists in both tables. However, the key difference is that it is a primary key in the owner table. This means it must be unique in each row of the table. Yet it might be duplicated in the vehicle table, because the same owner might have multiple vehicles. This means that the owner ID in the vehicle table is a suitable choice for a foreign key that connects the vehicle table with the owner table as shown in the following ERD:



This diagram shows the cross-reference between the two tables. The relationship is one-to-many, where each owner may own many Vehicles.

a. Create the relationship using the **"ALTER"** command to change the structure of the vehicle table, the "**ADD**" command to define the owner ID as the foreign key, and the "**REFERENCES**" keyword to reference it with the primary key in the owner table.
b. Show the new structure of the vehicle table: